BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021 • Paper Code: PR XI (01) 21

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A (Objective) — Model Paper

Time Allowed: 20 Minutes | **Maximum Marks:** 18

Note: Each question has four options. Choose the correct option. (The printed instructions for this section are in Urdu; given here in English.) CAVEAT: the objective sheet in the source is headed "Model Paper · Physics Paper – XI" (no year) — it is NOT the board's genuine 2021 objective sheet; kept as supplied.

Q1. Which of the given is least sub multiple?

(A) Micro
(B) Pico
(C) Femto
(D) Giga

Q2. One horse power is equal to .....

(A) 764 watts
(B) 756 watts
(C) 746 watts
(D) 784 watts

Q3. The no of significant digit in the number 0.000234 is .....

(A) 3
(B) 4
(C) 7
(D) 2

Q4. The magnitude of the Resultant of two forces is F. The magnitude of each force is also F. The angle b/w the forces must be .....

(A) 30°
(B) 60°
(C) 45°
(D) 120°

Q5. $(\hat{i} \times \hat{j}) \times \hat{k}$ = ...

(A) $\hat{i}$
(B) $\hat{j}$
(C) $\hat{k}$
(D) 0

Q6. If a.b = 1, a = 2 and b = 2 then the angle b/w the two vectors is .....

(A) 45°
(B) 30°
(C) 75°
(D) 60°

Q7. If A.B = 0 then A and B vectors are .....

(A) Perpendicular
(B) Parallel
(C) Normal
(D) None of these

Q8. A force of 3N and 4N acts on a body at the same time. Which of the given forces cannot be the resultant?

(A) 3.5N
(B) 0.3N
(C) 4N
(D) 5N

Q9. The power output of a lamp is 6 watt. How much energy does the lamp give out in 1 minute?

(A) 360J
(B) 12J
(C) 120J
(D) 720J

Q10. When an electron revolves around the nucleus in a circular path, the centripetal force is provided by .....

(A) Gravitational force
(B) Tension
(C) Columbic force
(D) Frictional force

Q11. The range of projectile is same for the two angles which are mutually .....

(A) Orthogonal
(B) Supplementary
(C) Complementary
(D) None of these

Q12. The work done by the body will be negative if the angle b/w force and displacement is .....

(A) 0°
(B) Less than 90°
(C) Greater than 90°
(D) 90°

Q13. The horizontal component of velocity of a body at highest point during projectile motion will be .....

(A) Maximum
(B) Minimum
(C) Zero
(D) Remain constant

Q14. If a body is moving in a circle of radius r. Then the angle b/w radius and tangential velocity is .....

(A) 30°
(B) 45°
(C) 90°
(D) 0°

Q15. A truck takes 80 seconds to travel the first 80m, and another 10 sec to travel a further 70m. What is the average speed?

(A) 0.004 m/s
(B) 1.6 m/s
(C) 4 m/s
(D) 5 m/s

Q16. Which of the given has the same dimensions?

(A) Inertia & torque
(B) Impulse & momentum
(C) Stress & strain
(D) Both A & B

Q17. Maximum work can be done in the process called .....

(A) Isothermal
(B) Adiabatic
(C) Isobaric
(D) Isochoric

Q18. The triple point of water is .....

(A) 273.16 C
(B) 273.16 K
(C) 372 K
(D) 0 C

Section-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 40

> Note: Note: Time allowed for Section-B and Section-C is 2 Hours and 40 minutes.

QII. Attempt any TEN parts. Each part carries FOUR marks.

1. Differentiate between precision and accuracy.
2. What are the practical examples of resonance?
3. Define torque with proper examples.
4. Write the necessary conditions of interference of waves.
5. Define elastic and inelastic collisions and give examples.
6. What is the Huygen's principle of light?
7. A soap bubble looks black when it bursts, why?
8. Derive relation for escape velocity.
9. Write the first and 2nd law of thermodynamics.
10. Define and explain Newton's 2nd law in terms of angular momentum.
11. How equation of continuity is based on the law of conservation of mass?
12. Explain work energy principle.

13. Derive the Stoke's Law for an ideal fluid.

Section-C

Time Allowed: 2 Hours 40 Minutes	**Maximum Marks:** 27

Note: Note: Time allowed for Section-B and Section-C is 2 Hours and 40 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII.

(a) [5] What is heat engine? Derive relation for the efficiency of a heat engine.

(b) [4] Calculate the change in entropy when 10 kg of water is heated from 90 °C to 100 °C. (specific heat of water is 4180 J mole−1k−1).

QIV.

(a) [5] Explain the experimental arrangement for the Young's double slits experiment for the interference of light. What are the important relations derived from this experiment about interference of light?

(b) [4] The speed of sound in air at 0 °C is 332 ms−1. What will be the speed of sound at 22 °C.

QV.

(a) [4] Find a relation for the terminal velocity of a spherical object falling in a resistive fluid.

(b) [5] Find the amplitude, frequency and time period of an object oscillating at the end of a spring, if the equation for its position at any instant (t) is given by: x = 0.25 cos(π/8)t. Find also the displacement of the object after 2.0 s.

QVI.

(a) [5] Find relation for the absolute P.E for a particle of mass m in the earth gravitational field.

(b) [4] A string 1 m long is used to whirl a 100 g stone in a horizontal circle at a speed of 2 m/s. Find the tension in the string.